CURRICULUM VITAE
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EDUCATION

2009 B.A. (Cum Laude); University of Washington; Seattle, WA, U.S.A. (*Anthropology, certificate in Medical Anthropology and Global Health, minor in African Studies*)

2011 M.A.; Pennsylvania State University; State College, PA, U.S.A. (*Anthropology and Demography*)

2014 Ph.D.; Pennsylvania State University; State College, PA, U.S.A. (*Anthropology and Demography*)

ACADEMIC APPOINTMENTS

March 2014 – June 2017
Postdoctoral Researcher and GeoUnit Supervisor: Shoklo Malaria Research Unit, Mahidol Oxford Tropical Medicine Research Unit, Mae Sot, Tak Province, Thailand

November 2020 – June 2022
Visiting Professor in Epidemiology; Faculty of Tropical Medicine, Mahidol University; Bangkok, Thailand

July 2017 – June 2023
Assistant Professor: Program in Public Health; Department of Population Health & Disease Prevention; University of California, Irvine

April 2019 – present
Assistant Professor (courtesy appointment): Department of Epidemiology & Biostatistics; School of Medicine; University of California, Irvine

July 2023 – present
Associate Professor (with tenure)**:** Program in Public Health; Department of Population Health & Disease Prevention; University of California, Irvine

PUBLICATIONS ([+] indicates corresponding author; [#] indicates co-first authorship; [!] indicates current or former mentee, student, or postdoctoral researcher)
N = 98; From Google Scholar: Citations = 3,233; H-index = 35

Peer reviewed papers:

2012

1 Cui L., Yan G., Sattabongkot J., Chen B., Cao Y., Fan Q., **Parker D.M.**, Sirichaisinthop J., Su X.Z., Yang H., Yang Z., Wang B., Zhou G. (2012). Challenges and prospects for malaria elimination in the Greater Mekong Subregion. Acta Tropica, 121(3): 240–245. https://doi.org/10.1016/j.actatropica.2011.04.006

2 Cui L., Yan G., Sattabongkot J., Cao Y., Chen B., Chen X., Fan Q., Fang Q., Jongwutiwes S., **Parker D.M.**, Sirichaisinthop J., Kyaw M.P., Su X.Z., Yang H., Yang Z., Wang B., Xu J., Zheng B., Zhong D., Zhou G.(2012). Malaria in the Greater Mekong Subregion: heterogeneity and complexity. Acta Tropica, 121(3): 227–239. https://doi.org/10.1016/j.actatropica.2011.02.016

3 Cui L., Wang Z., Jiang H., **Parker D.M.**, Wang H., Su X.Z., Cui L. (2012). Lack of association of the S769N mutation in *Plasmodium falciparum* SERCA (PfATP6) with resistance to artemisinins. Antimicrobial Agents and Chemotherapy, 56(5): 2546–2552. https://doi.org/10.1128/AAC.05943-11

4 [#]Wang Z., [#]**Parker D.M.**, Meng H., Wu L., Li J., Zhao Z., Zhang R., Fan Q., Wang H., Cui L., Yang Z. (2012). In vitro sensitivity of *Plasmodium falciparum* from China-Myanmar border area to major ACT drugs and polymorphisms in potential target genes. PLOS ONE, 7(5): e30927. https://doi.org/10.1371/journal.pone.0030927

5 [+]**Parker D.M.**, Zavortink T.J., Billo T.J., Valdez U., Edwards J.S. (2012). Mosquitoes and other arthropod macro fauna associated with tank bromeliads in a Peruvian cloud forest. Journal of the American Mosquito Control Association, 28(1): 45–46. https://doi.org/10.2987/11-6189.1

6 [+]**Parker D.M.**, Holman D. (2012). Event history analysis of dengue fever epidemic and inter-epidemic spells in Barbados, Brazil, and Thailand. International Journal of Infectious Diseases, 16(11): e793–e798. https://doi.org/10.1016/j.ijid.2012.07.002

7 [+]**Parker D.M.**, Lerdprom R., Srisatjarak W., Yan G., Sattabongkot J., Wood J., Sirichaisinthop J., Cui L. (2012). Longitudinal in vitro surveillance of *Plasmodium falciparum* sensitivity to common anti-malarials in Thailand between 1994 and 2010. Malaria Journal, 11(1): 290. https://doi.org/10.1186/1475-2875-11-290

2013

8 Weiss K.M., **Parker D.M.** (2013). Will you stop bugging me? Malaria and the evolutionary challenge that won't go away. Evolutionary Anthropology, 22(2): 46–51. https://doi.org/10.1002/evan.21340

9 Matthews S.A., **Parker D.M.** (2013). Progress in Spatial Demography. Demographic Research, 28: 271–312.

10 Yuan L., Zhao H., Wu L., Li X., **Parker D.M.**, Xu S., Zhao Y., Feng G., Wang Y., Yan G., Fan Q., Yang Z., Cui L. (2013). *Plasmodium falciparum* populations from northeastern Myanmar display high levels of genetic diversity at multiple antigenic loci. Acta Tropica, 125(1): 53–59. https://doi.org/10.1016/j.actatropica.2012.09.008

11 Miao M., Wang Z., Yang Z., Yuan L., **Parker D.M.**, Putaporntip C., Jongwutiwes S., Xangsayarath P., Pongvongsa T., Moji H., et al (2013). Genetic diversity and lack of artemisinin selection signature on the *Plasmodium falciparum* ATP6 in the Greater Mekong Subregion. PLOS ONE, 8(3): e59192. https://doi.org/10.1371/journal.pone.0059192

12 Miao J., Fan Q., **Parker D.M.**, Li X., Li J., Cui L. (2013). Puf mediates translation repression of transmission-blocking vaccine candidates in malaria parasites. PLOS Pathogens, 9(4): e1003268. https://doi.org/10.1371/journal.ppat.1003268

13 Miao J., Lawrence M., Jeffers V., Zhao F., **Parker D.M.**, Ge Y., Sullivan W. J., Cui L. (2013). Extensive lysine acetylation occurs in evolutionarily conserved metabolic pathways and parasite-specific functions during *Plasmodium falciparum* intraerythrocytic development. Molecular Microbiology, 89(4): 660–675. https://doi.org/10.1111/mmi.12303

14 Miao J., Wang Z., Liu M., **Parker D.M.**, Li X., Chen X., Cui L. (2013). *Plasmodium falciparum*: Generation of pure gametocyte culture by heparin treatment. Experimental Parasitology, 135(3): 541–545. https://doi.org/10.1016/j.exppara.2013.09.010

15 Li N., **Parker D.M.**, Yang Z., Fan Q., Zhou G., Ai G., Duan J., Lee M.C., Yan G., Matthews S.A., Cui L., Wang Y. (2013). Risk factors associated with slide positivity among febrile patients in a conflict zone of north-eastern Myanmar along the China-Myanmar border. Malaria Journal, 12(1): 361. https://doi.org/10.1186/1475-2875-12-361

2014

16 [+]**Parker D.M.**, Wood J.W., Tomita S., DeWitte S., Jennings J., Cui L. (2014). Household ecology and out-migration among ethnic Karen along the Thai-Myanmar border. Demographic Research, 30(39): 1129–1156.

17 Wang Z., Liu M., Liang X., Siriwat S., Li X., Chen X., **Parker D.M.**, Miao J., Cui L. (2014). A flow cytometry-based quantitative drug sensitivity assay for all *Plasmodium falciparum* gametocyte stages. PLOS ONE, 9(4): e93825. https://doi.org/10.1371/journal.pone.0093825

18 Li P., Zhao Z., Wang Y., Xing H., **Parker D.M.**, Yang Z., Baum E., Li W., Sattabongkot J., Sirichaisinthop J., Li S., Yan G., Cui L., Fan Q. (2014). Nested PCR detection of

malaria directly using blood filter paper samples from epidemiological surveys. Malaria Journal, 13(1): 175. https://doi.org/10.1186/1475-2875-13-175

19 He X., Yan J., Zhu X., Wang Q., Pang W., Qi Z., Wang M., Luo E., **Parker D.M.**, Cantorna M.T., et al. (2014). Vitamin D inhibits the occurrence of experimental cerebral malaria in mice by suppressing the host inflammatory response. The Journal of Immunology, 193(3): 1314–1323. https://doi.org/10.4049/jimmunol.1400089

20 Bancone G., Chu C.S., Somsakchaicharoen R., Chowwiwat N., **Parker D.M.**, Charunwatthana P., White N.J., Nosten F.H. (2014). Characterization of G6PD genotypes and phenotypes on the Northwestern Thailand-Myanmar border. PLOS ONE, 9(12): e116063. https://doi.org/10.1371/journal.pone.0116063

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21 Yuan L., Wang Y., **Parker D.M.**, Gupta B., Yang Z., Liu H., Fan Q., Cao Y., Xiao Y., Lee M.C., et al (2015). Therapeutic responses of *Plasmodium vivax* malaria to chloroquine and primaquine treatment in northeastern Myanmar. Antimicrobial Agents and Chemotherapy, 59(2): 1230–1235. https://doi.org/10.1128/AAC.04270-14

22 Tomita S., **Parker D.M.**, Jennings J., Wood J. (2015). Household demography and early childhood mortality in a rice-farming village in Northern Laos. PLOS ONE, 10(3): e0119191. https://doi.org/10.1371/journal.pone.0119191

23 **+Parker D.M.**, Matthews S.A., Yan G., Zhou G., Lee M.C., Sirichaisinthop J., Kiattibutr K., Fan Q., Li P., Sattabongkot J., Cui L. (2015). Microgeography and molecular epidemiology of malaria at the Thailand-Myanmar border in the malaria pre-elimination phase. Malaria Journal, 14: 198. https://doi.org/10.1186/s12936-015-0712-5

24 Moss W.J., Dorsey G., Mueller I., Laufer M.K., Krogstad D.J., Vinetz J.M., Guzman M., Rosas-Aguirre A.M., Herrera S., Arevalo-Herrera M., Chery L., Kumar A., Mohapatra P., Ramanathapuram L., Srivastava H., Cui L., Zhou G., **Parker D.M.**, Nankabirwa J., Kazura J. (2015). Malaria epidemiology and control within the international centers of excellence for malaria research. The American Journal of Tropical Medicine and Hygiene, 93(3 Suppl): 5–15. https://doi.org/10.4269/ajtmh.15-0006

25 Wang Y., Yang Z., Yuan L., Zhou G., **Parker D.M.**, Lee M.C., Yan G., Fan Q., Xiao Y., Cao Y., Cui L. (2015). Clinical efficacy of dihydroartemisinin–piperaquine for the treatment of uncomplicated *Plasmodium falciparum* malaria at the China–Myanmar border. The American Journal of Tropical Medicine and Hygiene, 93(3): 577–583. https://doi.org/10.4269/ajtmh.15-0029

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27 Kaji A., **Parker D.M.**, Chu C.S., Thayatkawin W., Suelaor J., Charatrueangrongkun R., Salathibuppha K., Nosten F.H., McGready R. (2016). Immunization coverage in migrant school children along the Thailand-Myanmar border. Journal of Immigrant and Minority Health, 18(5): 1038–1045. https://doi.org/10.1007/s10903-015-0294-x

28 Landier J., **Parker D.M.**, Thu A.M., Carrara V.I., Lwin K.M., Bonnington C.A., Pukrittayakamee S., Delmas G., Nosten F.H. (2016). The role of early detection and treatment in malaria elimination. Malaria Journal, 15: 363. https://doi.org/10.1186/s12936-016-1399-y

29 Gupta B., **Parker D.M.**, Fan Q., Reddy B.N., Yan G., Sattabongkot J., Cui L. (2016). Microgeographically diverse *Plasmodium vivax* populations at the Thai-Myanmar border. Infection, Genetics and Evolution, 45: 341–346. https://doi.org/10.1016/j.meegid.2016.09.021

30 [+]**Parker D.M.**, Landier J., von Seidlein L., Dondorp A., White L.J, Hanboonkunupakarn B., Maude R.J., Nosten F.H. (2016). Limitations of malaria reactive case detection in an area of low and unstable transmission on the Myanmar–Thailand border. Malaria Journal, 15: 571. https://doi.org/10.1186/s12936-016-1631-9

2017

31 Ya-umphan P., Cerqueira D., **Parker D.M.**, Cottrell G., Poinsignon A., Remoue F., Brengues C., Chareonviriyaphap T., Nosten F., Corbel V. (2017). Use of an *Anopheles* salivary biomarker to assess malaria transmission risk along the Thailand-Myanmar border. The Journal of Infectious Diseases, 215(3): 396–404. https://doi.org/10.1093/infdis/jiw543

32 [#]Srikanok S., [+#]**Parker D.M.**, Parker A.L., Lee T., Min A.M., Ontuwong P., Tan S.O., Sirinonthachai S., McGready R. (2017). Empirical lessons regarding contraception in a protracted refugee setting: A descriptive study from Maela camp on the Thai-Myanmar border 1996–2015. PLOS ONE, 12(2): e0172007. https://doi.org/10.1371/journal.pone.0172007

33 [#]Thi S., [+#]**Parker D.M.**, Swe L., Pukrittayakamee S., Ling C., Amornpaisarnloet K., Vincenti-Delmas M., Nosten F. (2017). Migration histories of multidrug-resistant tuberculosis patients from the Thailand-Myanmar border, 2012–2014. The International Journal of Tuberculosis and Lung Disease, 21(7): 753–758. https://doi.org/10.5588/ijtld.16.0658

34 Thu A., Phyo A., Landier J., **Parker D.M.**, Nosten F. (2017). Combating multi-drug resistant *Plasmodium falciparum* malaria. The FEBS Journal, 284(16): 2569 – 2578. https://doi.org/10.1111/febs.14127

35 Beckerman S., Lizarralde M., Peluso D., Yvinec C., Harris N., **Parker D.M.**, Walker R., Hill K. (2017). Partible paternity, the secondary sex ratio, and a possible Trivers-Willard effect. Current Anthropology, 58(4): 540 – 543.

36 Kajeechiwa L., Thwin M.M., Nosten S., Tun S.W., **Parker D.M.**, von Seidlein L., Tangseefa D., Nosten F., Cheah P.Y. (2017). Community engagement for the rapid elimination of malaria: the case of Kayin State, Myanmar. Wellcome Open Research, 2:59. https://doi.org/10.12688/wellcomeopenres.12051.1

37 Landier J., Kajeechiwa L., Thwin M.M., **Parker D.M.**, Wiladphaingern J., Imwong M., Miotto O., Patumrat K., Duanguppama J., Cerqueira D., Malleret B., Rénia L., Nosten S., von Seidlein L., Ling C., Proux S., Corbel V., Simpson J.A., Dondorp A.M., White N.J., Nosten F.H. (2017). Safety and effectiveness of mass drug administration to accelerate

elimination of artemisinin-resistant falciparum malaria: A pilot trial in four villages of Eastern Myanmar. Wellcome Open Res, 2:81. https://doi.org/10.12688/wellcomeopenres.12240.1

38 [+]**Parker D.M.**, Tripura R., Peto T.J., Maude R.J., Nguon C., Chalk J., Sirithiranont P., Imwong M., von Seidlein L., White N.J., Dondorp A.M. (2017). A multi-level spatial analysis of clinical malaria and subclinical Plasmodium infections in Pailin Province, Cambodia. Heliyon, 3: e00447. https://doi.org/10.1016/j.heliyon.2017.e00447

39 Bonnington C.A., Phyo A.P., Ashley E., Imwong M., Sriprawat K., **Parker D.M.**, Proux S., White N.J., Nosten F. (2017). *Plasmodium falciparum* Kelch 13 mutations and treatment response in patients in Hpa-Pun District, Northern Kayin State, Myanmar. Malaria Journal, 16: 480. https://doi.org/10.1186/s12936-017-2128-x

40 [!]Tun S.T.T., von Seidlein L., Pongvongsa T., Mayxay M., Saralamba S., Kyaw S., Chanthavilay P., Celhay O., Nguye T.D., Tran T.N., **Parker D.M.**, Boni M.F., Dondorp A.M., White L.J. (2017). Towards malaria elimination in Savannakhet, Lao PDR: mathematical modelling driven strategy design. Malaria Journal, 16: 483. https://doi.org/10.1186/s12936-017-2130-3

41 [+#]**Parker D.M.**, [#]Landier J., Thu A.M., Lwin K.M., Delmas G., Nosten F., the Malaria Elimination Task Force Group. (2018). Scale up of a *Plasmodium falciparum* elimination program and surveillance system in Kayin State, Myanmar. Wellcome Open Research, 2(98). https://doi.org/10.12688/wellcomeopenres.12741.2

2018

42 [#]Landier J., [#]**Parker D.M.**, Thu A.M., Lwin K.M., Delmas G., Nosten F., the Malaria Elimination Task Force Group (2018). Effect of generalized access to early diagnosis and treatment and targeted mass drug administration on *Plasmodium falciparum* malaria in Eastern Myanmar: an observational study of a regional elimination programme. The Lancet, 391: 1916 - 1926. https://doi.org/10.1016/S0140-6736(18)30792-X

43 Ya-Umphan P., Cerqueria D., Cottrell G., **Parker D.M.**, Fowkes F., Nosten F., Corbel V. (2018). *Anopheles* salivary biomarker as a proxy for estimating *Plasmodium falciparum* malaria exposure on the Thailand-Myanmar border. American Journal of Tropical Medicine and Hygiene, 99(2): 350 – 356. https://doi.org/10.4269/ajtmh.18-0081

44 Parker A.L., [+]**Parker D.M.**, Zan B.N., Min A.M., Gilder M.E., Ringringulu M., Win E., Wiladphaingern J., Charunwatthana P., Nosten F., Lee S.J., McGready R. (2018). Trends and birth outcomes in adolescent refugees and migrants from the Thailand-Myanmar border, 1986 – 2016: an observational study. Wellcome Open Research, 3:62. https://doi.org/10.12688/wellcomeopenres.14613.1

45 Manning J.E., Oliveira F., **Parker D.M.**, Amaratunga C., Kong D., Man S., Sreng S., Lay S., Nang K., Kimsan S., Sokha L., Kamhawi S., Fay M.P., Suon S., Ruhl P., Ackerman H., Huy R., Wellems T.E., Valenzuela J.G., Leang R. (2018). The PAGODAS protocol: pediatric assessment group of dengue and *Aedes* saliva protocol to investigate vector-borne determinants of *Aedes*-transmitted arboviral infections in Cambodia. Parasites & Vectors 11(1):664. https://doi.org/10.1186/s13071-018-3224-7

2019

46 von Seidlein L., Peto T.J., Landier J., Nguyen T.N., Tripura R., Phommasone K., Pongvongsa T., Lwin K.M., Keereecharoen L., Kajeechiwa L., Thwin M.M., **Parker D.M.** Wiladphaingern J., Nosten S., Proux S., Corbel V., Tuong-Vy N., Phuc-Nhi T.L., Hung Son D., Huong-Thu P.N., Tuyen N.T.K., Tien N.T., Dong L.T., Hue D.V., Quang H.H., Nguon C., Davoeung C., Rekol H., Adhikari B., Henriques G., Phongmany P., Suangkanarat P., Jeeyapant A., Vihokhern B., van der Pluijm R., Lubell Y., White L.J., Aguas R., Promnarate C., Sirithiranont P., Malleret B., Renia L., Onsjo C., Chan X.H., Chalk J., Miotto O., Patumrat K., Chotivanich K., Hanboonkunupakarn B., Jittmala P., Kaehler N., Cheah P.Y., Pell C., Dhorda M., Imwong M., Snounou G., Mukaka M., Peerawaranun P., Lee S.J., Simpson J.A., Pukrittayakamee S., Singhasivanon P. Grobusch M.P., Cobelens F., Smithuis F., Newton P.N., Thwaites G.E., Day N.P.J., Mayxay M., Hien T.T., Nosten F.H., Dondorp A.M., White N.J. (2019). The impact of targeted malaria elimination with mass drug administrations on falciparum malaria in Southeast Asia: A cluster randomised trial. PLOS Medicine 16(2): e1002745. https://doi.org/10.1371/journal.pmed.1002745

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51 [+]**Parker D.M.** (2019). Humans, macaques, and malaria parasites in a shared and changing landscape. Lancet Planetary Health. 3(4): e157 – e158. https://doi.org/10.1016/S2542-5196(19)30061-0

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Contribution of asymptomatic Plasmodium infections to the transmission of malaria in Kayin State, Myanmar. (2019). The Journal of Infectious Diseases 219(9):1499-1509. https://doi.org/10.1093/infdis/jiy686

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53 Chu C.S., Carrara V.I., **Parker D.M.**, Proux S., Charunwatthana P., McGready R., Nosten F. (2020). Declining burden of *Plasmodium vivax* in a population in Northwestern Thailand from 1995 to 2016 before comprehensive primaquine prescription for radical cure. American Journal of Tropical Medicine and Hygiene. 102(1): 147-150. https://doi.org/10.4269/ajtmh.19-0496

54 Phommasone K., Leth F., Peto T.J., Landier J., Nguyen T.N., Tripura R., Pongvongsa T., Lwin K.M., Kajeechiwa L., Thwin M.M., **Parker D.M.**, Wiladphaingern J., Nosten S., Proux S., Nguon C., Davoeung C., Rekol H., Adhikari B., Promnarate C., Chotivanich K., Hanboonkunupakarn B., Jittmala P., Cheah P.Y., Dhorda M., Imwong M., Mukaka M., Peerawaranun P., Pukrittayakamee S., Newton P.N., Thwaites G.E., Day N.P.J., Mayxay M., Hien T.T., Nosten F.H., Cobelens F., Dondorp A.M., White N.J., von Seidlein L. (2020). Mass drug administrations with dihydroartemisinin-piperaquine and single low dose primaquine to eliminate *Plasmodium falciparum* have only a transient impact on *Plasmodium vivax*: findings from a randomised controlled trial. PLOS ONE. 15(2):e0228190. https://doi.org/10.1371/journal.pone.0228190

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transmission in Botswana using geospatial and genomic data – the Kopanyo Study. Emerging Infectious Diseases, 29(5), 977-987. https://doi.org/10.3201/eid2905.220796

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97. Bayer D., Goldstein I.H., Fintzi J., Lumbard K., Ricotta E., Warner S., Strich J.R., Chertow D.S., Busch L.M., **Parker D.M.**, Boden-Albala B., Chhuon R., Zahn M., Quick N., Dratch A., Minin V.M. Semi-parametric modeling of SARS-CoV-2 transmission using tests, cases, deaths, and seroprevalence data. Annals of Applied Statistics. [accepted Feb 15, 2024]

Chapters

2016 [+]**Parker D.M.** Human migration and spatial synchrony: spatial patterns in temporal trends. Pp. 325-351, Chapter 16 in *Recapturing space: new middle-range theory in spatial demography*, edited by Frank M. Howell, Jeremy R. Porter, and Stephen A. Matthews. Springer, New York, NY. https://doi.org/10.1007/978-3-319-22810-5_16

2023 [!]Glendening N., [!]Haileselassie W., [+]**Parker D.M.** A conceptual framework for understanding extractive settlements and disease: demography, environment, and epidemiology. Pp. 121 – 160, Chapter 5 in *Planetary health approaches to understand and control vector-borne diseases*, edited by Kimberly Fornace, Jan Conn, Maria Anice Mureb Sallum, Leonardo Suveges Moreira Chaves, and James Logan. Wageningen Academic, Leiden, The Netherlands https://doi.org/10.3920/9789004688650_007

GRANTS, CONTRACTS, RESEARCH AWARDS, AND CONSULTATIONS

External Funding

Active

2020 – 2024 National Institutes of Health/Fogarty International Center. U.S. Mentor. Project title: "Population biology of African malaria vectors." D43 Training Grant; FAIN: TW001505; PI: Guiyun Yan ($270,000 annual direct costs)

2021 - 2024 National Institute of Allergy and Infectious Disease. Subrecipient Principal Investigator. Project title: "Assessing the acceptability of malaria interventions: a mixed methods approach." U19 Clinical Trial Option; FAIN: U19AI089672; Subaward number: 6123-1247-12-D; Part of the Southeast Asian Center for Excellence in Malaria Research; PI: Liwang Cui ($294,324)

2023 – 2024 PacVec (Pacific Southwest Center of Excellence in Vector-Borne Diseases). Principal Investigator. Project title: "Geospatial surveillance and geo-targeted interventions for halting the spread of *Ae. aegypti* mosquitoes in San Bernardino County, California." Award number: A23-0750-S015 ($25,000)

2023 – 2024 Wellcome Trust. Subrecipient Principal Investigator. Project title: "Health of Refugees and Forcibly Displaced Communities."; Subaward number: 4500004783; Award Number: 601412. PI: Muhammad H. Zaman ($59,552)

Completed

2014 - 2017 Bill and Melinda Gates Foundation. Co-Investigator. "Targeted malaria elimination in Karen State, Myanmar." Award: OPP1117507. PI: François Nosten. ($6,326,573)

2017 - 2018 Thailand Regional Artemisinin Initiative, through the Global Fund to fight AIDS, Tuberculosis and Malaria. Co-Investigator. "GPS tracking of at-risk population movement in the Thai-Myanmar border region." PI: François Nosten. ($20,000)

2019 - 2021 Bill and Melinda Gates Foundation. Co-Principal Investigator. Project title: "Identification and characterization of vector-borne pathogens and vector exposures to define regional biomonitoring strategies and vector control efforts in Cambodia." Award: OPP1211806. PI: Jessica Manning. ($100,000)

2019 - 2021 National Institute of Allergy and Infectious Disease. Subrecipient Principal Investigator. Project title: "Assessing the acceptability of malaria interventions: a

mixed methods approach." U19 Clinical Trial Option; FAIN: U19AI089672; Subaward number: 6123-1247-00-D; Part of the Southeast Asian Center for Excellence in Malaria Research ($198,000)

2020 – 2021 Federal Emergency Management Agency. Co-Investigator. "Population-wide surveillance of COVID-19 antibodies in Orange County." In collaboration with Orange County Health Care Agency; contract: MA-042-20011978. PI: Tim Bruckner (mPI Bernadette Boden-Albala). ($1,700,000)

2020 – 2021 Federal Emergency Management Agency. Co-Investigator. "Seroprevalence surveillance and predictive modeling for COVID-19 in Santa Ana, California." contract: A-2020-158-36. PI: Bernadette Boden-Albala). ($2,960,145)

2020 - 2021 Federal Emergency Management Agency. Principal Investigator. "Health equity contact tracing." In collaboration with Orange County Health Care Agency; contract: MA-042-21010038. ($217,045)

2021 – 2023 Bill and Melinda Gates Foundation. Co-Principal Investigator. Project title: "Pathogen metagenomics in the Mekong: A multi-faceted approach to improve public health." Award: INV-028123. PI: Jessica Manning. ($993,567)

Internal Funding

2008 Fritz Grant for International Study, University of Washington

2008 Stewart Alva Nereim Scholarship ($1,000)

2008 Myra A. and Suzanne Johnson Hansen Memorial Scholarship ($4,332)

2009 IGERT Program in Evolutionary Modeling (IPEM) Fellowship, University of Washington, (declined)

2010 RGSO travel grant, Penn State University ($400)

2011 Paul T. Baker Research Travel Fund in Human Biology and Anthropology, Department of Anthropology, Pennsylvania State University ($2,000)

2011 RGSO travel grant, Pennsylvania State University ($400)

2011 Tag Along Fund, College of Agricultural Sciences, Pennsylvania State University ($750)

2012 Paul T. Baker Research Travel Fund in Human Biology and Anthropology, Department of Anthropology, Pennsylvania State University ($416)

2012 RGSO travel grant, Pennsylvania State University ($500)

2012 Hill Fellowship Post-Comprehensive Examination Award, Department of Anthropology, Pennsylvania State University ($2,854)

2013 Donald E. Harvey Graduate Scholarship, College of Liberal Arts, Pennsylvania State University ($2,000)

2013 Matson Museum Ceramics Award, Department of Anthropology, Pennsylvania State University ($4,600)

2013 Hill Fellowship Post-Comprehensive Examination Award, Department of Anthropology, Pennsylvania State University ($6,120)

2013 William T. Sanders award in Anthropology, Pennsylvania State University ($1,000)

2013 Research and Graduate Studies (RGSO) travel grant, Pennsylvania State University ($457)

2013 Demography Program Travel Grant, Pennsylvania State University ($500)

2014 Shultz Family Graduate Fellowship, College of Liberal Arts, Pennsylvania State University (declined) ($4,000)

2018 Faculty Research Grant, Council on Research, Computing, and Libraries (CORCL), University of California, Irvine ($1,500)

2019 Faculty Seed Grant; Center for Population, Inequality, and Policy (CPIP); University of California, Irvine ($4,442)

2019 Faculty Research Grant, Council on Research, Computing, and Libraries (CORCL), University of California, Irvine ($1,500)

2020 Faculty Research Grant, Council on Research, Computing, and Libraries (CORCL), University of California, Irvine ($2,000)

2021 Faculty Research Grant, Council on Research, Computing, and Libraries (CORCL), University of California, Irvine ($2,000)

2022 Faculty Research Grant, Council on Research, Computing, and Libraries (CORCL), University of California, Irvine ($946)

2022 Seed funding through the “Investing to Develop Center-Scale Multidisciplinary Convergence Research Programs” initiative. Project title: "New Data Streams and Predictive Analytics for Pandemic Preparedness" (Aug 1, 2022 - July 30, 2024) PI: Vladimir Minin; role: Co-Investigator ($148,005)

2023 Faculty Research Grant, Council on Research, Computing, and Libraries (CORCL), University of California, Irvine ($4,929)

2024 Surgery Seed Grant, School of Medicine, University of California, Irvine. Project title: “Evaluating Access to Neurosurgical Care in Ethiopia” (January – December 2024) PI: Alliya Qazi; role: Co-Investigator ($7,500)

Consultations

2014 Community Partners International Myanmar. Consultant. Mapping and GIS for malaria elimination.

2015 Save the Children Nepal (for the Global Fund to fight AIDS, Tuberculosis and Malaria). Consultant. Mapping and GIS for malaria health services and elimination.

2020 – 2021 Expert witness for a legal case about COVID-19 transmission in San Quentin Prison, Marin County, California.

HONORS & AWARDS (selected)

2009 Curtis Wienker award for Best Biocultural Anthropology Essay, University of Washington

2010 Juan Comas Award for best student paper, American Association of Physical Anthropologists Annual Meeting, Albuquerque, NM

2020 Dean's Honoree for Public Health Celebration of Teaching, University of California, Irvine (https://www.youtube.com/watch?v=WIT0VAhF-vM)

2020 Inducted into UCI's Delta Rho chapter of the Delta Omega Honorary Society in Public Health

2020 Distinguished panelist for "The Future of the COVID Curve" at the Orange County Forum (https://www.youtube.com/watch?v=DY34AukTnII)

2021 Panelist for the "Madeline Martin Swinden Memorial Lecture" for the Samueli College of Health Sciences, University of California, Irvine

2021 Chartered Geographer and Fellow, Royal Geographical Society, London, U.K.

INVITED TALKS, PANELS, and SYMPOSIA

1. Human spatial ecology and residual transmission of malaria. Invited speaker: Entomology and Vector Control for Malaria Elimination Regional Meeting & Workshop; Asia Pacific Malaria Elimination Network; Bangkok, Thailand. 2016: https://apmen.org/events/7-10-november-2016

2. The importance of human population movement: Scale and frequency in infectious diseases dynamics. Invited speaker: Infectious Routes: Epidemics and migration in Asia. Hong Kong University, Hong Kong. 2017.

3. Scale up and preliminary results from a *P. falciparum* malaria elimination program in Eastern Kayin State, Myanmar. Invited speaker: Center for Malaria Research at Penn State, Pennsylvania State University, University Park, PA. 2017.

4. Who benefits from new technology? Perspectives from the Thai-Myanmar border. Invited speaker: Phones, drones and diseases. Hong Kong University, Hong Kong. 2018.

5. Geospatial co-variation in entomological indices in malaria and dengue systems within the Mekong. Invited speaker: Defining vector research priorities in the Greater Mekong Sub-Region. Phnom Penh, Cambodia. 2019.

6. Town hall discussion with Senator Ling Ling Chang on contact tracing for COVID-19 (Senator for the 29th District of California), July 8, 2020: https://voiceofoc.org/2020/06/oc-partners-with-community-group-to-reduce-coronavirus-impacts-on-anaheim-santa-ana/

7. Panelist on OC Forum: The Future of: The COVID Curve (July 9, 2020): https://www.youtube.com/watch?v=DY34AukTnII

8. Presentation to the Anaheim Unified High School District and the Anaheim Elementary School District Boards (both in Orange County, California) on whether or not school should be in-person (both voted unanimously to open the school year remotely, July 16, 2020): https://www.ocregister.com/2020/07/16/two-anaheim-school-districts-wont-reopen-campuses-this-fall/

9. Multilevel effects of targeted mass drug administration for elimination of *P. falciparum* malaria on the Thailand-Myanmar border. Invited speaker for a session on: "Malaria parasite epidemiology and population dynamics during elimination" at the Joint International Tropical Medicine Meeting. Bangkok, Thailand (S3, virtual). 2020.

10. Reported acceptability and feasibility of mass administration of primaquine in field sites in Thailand and Myanmar. Invited speaker for a session on "Community and health system research for malaria elimination" at the Joint International Tropical Medicine Meeting. Bangkok, Thailand (S11, virtual). 2020.

11. Updates on a *P. falciparum* elimination campaign in Kayin State, Myanmar; a TB program on the Thai-Myanmar border; and spatial epidemiology of COVID-19 in Orange County, California (USA). Invited speaker: Texas Biomedical Research Institute (virtual). 2021. https://www.newsbreak.com/videos/2229909465688/texas-biomed-seminar-series-dr-daniel-m-parker-university-of-california-irvine

12. Experience Sharing: Using qualitative methods to enhance epidemiological research. Invited speaker, along with Bipin Adhikari: Mahidol University. Bangkok, Thailand (virtual). 2021

13. Mapping mosquito spit: a novel way to assess the space-time risk of mosquito borne diseases. Invited speaker for the Epidemiology Seminar Series at the Department of Epidemiology in the School of Public Health at the University of Michigan. (virtual). 2021

14. Geospatial analyses of *Aedes aegypti* antibody intensity and its relationship to disease status. Invited speaker for IVCC (Innovative Vector Control Consortium) Tech Bites. (virtual). 2021

15. A spatial analysis of exposure to *Aedes* mosquitoes in an peri-urban area endemic to dengue, chikungunya, and Zika viruses. Invited speaker for the Infectious Diseases Grand Rounds in the School of Medicine at the University of California, Irvine. (virtual). 2022

16. Mapping antibody responses to mosquito saliva for spatial epidemiologic analyses. Invited speaker for The GeoHealth Hour at the University of Twente, Netherlands. (virtual). 2022

17. Spatial analysis of exposure to *Aedes* mosquitoes in a dengue-hyperendemic setting in Cambodia. Invited speaker at the Faculty of Tropical Medicine, Mahidol University, Bangkok, Thailand. 2022.

18. Application of anthropology and demography for building a malaria elimination program. Guest lecture for an Anthropology course at the University of South Dakota, 2023.

19. Social Epidemiology of Vector Borne Diseases, Prevention and Control. S19 at the Joint International Tropical Medicine Meeting in Bangkok, Thailand. Co-chaired with Suparat Phuanukoonnon. December 14, 2023.

20. Contrasting environmental factors in scrub and murine typhus infections in Lao PDR (2003 – 2017). Invited lecture at the NIAID (National Institute of Allergy and Infectious Disease) International Center of Excellence in Research (ICER), Phnom Penh, Cambodia. 2023.

21. A spatio-temporal analysis of scrub typhus cases in Lao PDR w/ murine typhus as a comparison. Invited talk for the American Society for Rickettsiology 'Spotlight on Rickettsia'. (virtual) 2024.

22. Panelist for Dean's Distinguished Lecture Series "Approaches to Population Health, Locally and Globally." (February 28, 2024).

PEER-REVIEWED CONFERENCE PRESENTATIONS (* indicates presenter)

1. ***Parker D.**, Medina C. Comparison of chiroptera and arthropod diversity in the Amazon Basin and the Andean Cloud Forest. *Centro de Educacion, Ciencia, y Conservacion Tambopata (CECCOT), Puerto Maldonado, Peru.* 2008. [talk]

2. ***Parker D.** A survey of the arthropod populations within the phytotelmata of tank bromeliads in the cloud forests of the Andes. *UW Undergraduate Research Symposium.* 2009. [poster]

3. ***Parker D.**, Curran S. The meanings of destination enclaves in origin communities and their consequences for migrant networks between origin and destination communities. *Population Association of America, Dallas, TX. P3-58.* 2010. [poster]

4. ***Parker D.**, Holman D. An event history analysis of dengue fever outbreaks in eight different endemic regions. *American Association of Physical Anthropologists, Albuquerque, NM. 141(S50):pp. 184-185.* 2010. [poster]

5. ***Parker D.**, Jennings J. Wood J. Sex and age-specific trends in the loss of life from tuberculosis in Sanday and Westray, Northern Orkney (Scotland), 1861-1961. *American Association of Physical Anthropologists, Minneapolis, MN. 144(S52):p. 234.* 2011. [poster]

6. *__Parker D.__, Lerdprom R., Srisatjarak W., Yan G., Prachumsri J., Cui L., Sirichaisinthop J. Antimalarial drug resistance surveillance in Thailand: 1994 – 2010. *American Society of Tropical Medicine and Hygiene, Philadelphia, PA. 85(6 Suppl): LB-2286.* 2011. [poster]

7. *__Parker D.__, Jennings J. Wood J. Transitions in mortality and documented causes of death: North Orkney (1855 – 1955). *American Association of Physical Anthropologists, Portland, OR. 147(S54):pp. 232-233.* 2012. [poster]

8. *Jennings J., __Parker, D__., Wood J. The influence of environmental and household factors on mortality in North Orkney, 1851 – 1961. *Population Association of America, San Francisco, CA. P2-32.* 2012. [poster]

9. *__Parker D.__, Matthews S., Srisatjarak W., Boonpan P., Lerdprom R., Lee M., Yan G., Sattabongkot J., Cui L., Sirichaisinthop J. A spatio-temporal analysis of *Plasmodium falciparum* and *P. vivax* infections and treatment seeking behavior in Tha Song Yang District, Tak Province, Thailand: 2008 - 2011. *American Society of Tropical Medicine and Hygiene, Atlanta, GA. 877.* 2012. [poster]

10. *__Parker D.M.__, Wood J.W., Tomita S., DeWitte S., Sirichaisinthop J., Cui L. Household level predictors and migration: a micro-demographic study of Karen villages along the Thai-Myanmar border. *Population Association of America, New Orleans, LA. P3-76.* 2013. [poster]

11. *__Parker D.M.__, Wood J.W., Tomita S., DeWitte S., Sirichaisinthop J., Cui L. A sensitivity analysis of the impact of household consumer-producer ratios on the hazard of out-migration among the Karen. *American Association of Physical Anthropologists, Knoxville, TN. 150(S56): pp. 216-217.* 2013. [talk]

12. __Parker D.M.__, Matthews S.A., *Sattabongkot J., Sirichaisinthop J., Kiattibutr K., Yan G., Zhou G., Lee M., Fan Q., Li P., Cui L. Malaria in borderlands: A case study from the Thailand-Myanmar border. *American Society of Tropical Medicine and Hygiene, New Orleans, LA. 941.* 2014. [poster]

13. *Ya-umphan P., __Parker D.M.__, Cerqueira D., Cottrell G., Poinsignon A., Remoue F., Charoenviriyaphap T, Nosten, F., Corbel V. Human antibody response to *Anopheles* salivary peptide as a biomarker to assess the risk of malaria transmission in the Thai-Myanmar border**.** *Joint International Tropical Medicine Meeting, Bangkok, Thailand*. 2015. [poster]

14. *__Parker D.M.__, Landier J., Wiladpaingern J., Kereecharoen L., Maelankiri L., Thwin M.M., Nosten F.H. Complex spatial patterns in Southeast Asian falciparum malaria transmission mean that reactive case detection is unlikely to halt transmission. *Joint International Tropical Medicine Meeting, Bangkok, Thailand*. 2015. [talk]

15. Thi S.S., *__Parker D.M.__, Swe L.L., Ling C., Nosten F.H. Migrants with multi-drug resistant tuberculosis. *XXV SMRU Border Meeting: Border Health, Malaria and Infectious Disease Forum, Mae Sot, Thailand.* 2015. [talk]

16. *Bonnington C., **Parker D.M.**, Landier J., Nosten, F. Practicalities of malaria elimination in Eastern Burma. *Oxford Tropical Network Conference, Siem Reap, Cambodia.* 2015. [poster]

17. ***Parker D.M.** Spatial analysis of the re-emergence of clinical and asymptomatic *Plasmodium falciparum* in a low-uptake targeted chemo-elimination study site. *Oxford Tropical Network Conference, Siem Reap, Cambodia.* 2015. [talk]

18. *Swe Lei Lei, **Parker D.M.**, Nosten S., Charunwatthana P., Vincenti-Delmas M. Putting life on hold: A clinic becomes a community for tuberculosis patients. *Joint International Tropical Medicine Meeting, Bangkok, Thailand*. 2015. [poster]

19. *Taylor A., Cerqueira G.C., Nkhoma S.C., Nair S., Cheeseman I.H., McDew-White M., Phyo A.P., **Parker D.M.**, Nosten F., Anderson T.J.C., Wesolowski A., Engø-Monsen K., Neafsey D.E., Buckee C. Malaria parasite population structure and implied human mobility in Northwest Thailand. *American Society of Tropical Medicine and Hygiene, Atlanta, GA. 1580.* 2016. [poster]

20. ***Parker D.M.,** Landier J., Thu A.M., Kajeechiwa L., Twin M.M., Soe T.N., Proux S., Tun S.W., Min M.C., Moo K.L., Lwin K.M., Khin S.D., Marta E., Delmas G., Nosten F. Geographic distribution and spatial clustering of submicroscopic malaria in Kayin State, Myanmar. *American Society of Tropical Medicine and Hygiene, Atlanta, GA. 922.* 2016. [poster]

21. Landier J., ***Parker D.M.**, Thu A.M., Kajeechiwa L., Twin M.M., Soe T.N., Proux S., Tun S.W., Min M.C., Moo K.L., Lwin K.M., Khin S.D., Marta E., Delmas G., Nosten F. Relative contribution of generalized early diagnosis and treatment and of targeted mass treatment to elimination of *Plasmodium falciparum* malaria in Eastern Myanmar. *American Society of Tropical Medicine and Hygiene, Atlanta, GA. 1217.* 2016. [talk]

22. *Landier J., Santirad A., Wanachaloemlap C., B'let S., Kittitawee K., Konghahong K., Christensen P., Haohankhunnatham W., Archasuksan L., Wongaeka M., Raksuansak J., Das S., Jang I.K., Wiladphaingern J., **Parker D.M.**, Thu A.M., Ling C., Proux S., Domingo G.J., Delmas G., Nosten F. A new high sensitivity rapid diagnostic test for *Plasmodium falciparum* detection in an elimination setting: individual diagnosis of infection and measurement of prevalence. *American Society of Tropical Medicine and Hygiene, Baltimore, MD. 967.* 2017. [poster]

23. *Tomita S., **Parker D.M.** Household demography and land-use in a rice-farming village in Laos from 1971 to 2013. *American Association of Physical Anthropologists, New Orleans, LA. S56.* 2017. [poster]

24. *Pan-ngum W., Boondam J., White L.J., **Parker D.M.** Cross-border malaria and new challenges to Thai national malaria control programme. *British Society for Parasitology, City of Dundee, UK. A12865.* 2017. [talk]

25. *Roberts T., Rattanavong S., Elliot I., Phommasone K., Mayxay M., Chansamouth V., Homsana A., Robinson M., Blacksell S., **Parker D.M.**, Newton P.N. Come rain or come shine. Differential spatiotemporal dynamics of scrub typhus and murine typhus in the Lao PDR. *American Society of Tropical Medicine and Hygiene, New Orleans, LA. 754.* 2018. [talk]

26. Rattanavong S., Dubot-Pérès A., Mayxay M., Vongsouvath M., Lee S.J. Cappelle J., Newton P.N., ***Parker D.M.** A spatial analysis of central nervous system infections in Lao PDR. *American Society of Tropical Medicine and Hygiene, New Orleans, LA. LB-5125.* 2018. [poster]

27. *Bipin Adhikari B., Phommasone K., Pongvosa T., Koummarasy P., Soundala X., Henriques G., Sirithiranont P., **Parker D.M.**, von Seidlein L., White N.J., Day N.P.J., Dondorp A.M., Newton P.N., Cheah P.Y., Pell C., Mayxay M. A mixed method study of health seeking behavior for febrile illnesses and its implications for malaria control and elimination in Savannakhet Province, Lao PDR. *American Society of Tropical Medicine and Hygiene, New Orleans, LA. 400.* 2018. [poster]

28. Leang R., **Parker D.M.**, Kong D., Man S., Sreng S., Lay S., Nang K., Kamhawi S., Fay M., Amaro-Carambot E., Whitehead S., Suon S., Huch C., Huy R., Wellems T.E., Valenzuela J.G., *Manning J.E. Update from PAGODAS: Pediatric Assessment Group Of Dengue and *Aedes* Saliva to investigate vector-borne determinants of *Aedes*-transmitted arboviral infections in Cambodia. *American Society of Tropical Medicine and Hygiene, National Harbor, MD. 9.* 2019. [talk]

29. *Li X, Arya GA, Reyes A, Thu AM, Delmas G, **Parker DM**, Lwin KM, Sriprawat K, Nosten F, Anderson T. Spatial analysis of parasite population genomics during malaria elimination efforts in Eastern Myanmar. *American Society of Tropical Medicine and Hygiene, National Harbor, MD. 1938.* 2019. [talk]

30. *Roberts T, **Parker DM**, Butlerys P, Phommasone K, Chansamouth V, Vongsouvath M, Rattanavong S, Elliot I, Robinson M, Blacksell S, Mayxay M, Newton PN. Differential spatiotemporal dynamics of scrub typhus and murine typhus in the Lao PDR. *Asia-Pacific Rickettsia Conference, Chiang Rai, Thailand. 37.* 2019. [talk]

31. *Kong D., Lay S., Man S., **Parker D.M.**, Sreng S. Chea S., Suon S., Huch C., Huy R., Oliveira F., Leang R., Ayhong V., Tato C., DeRisi J., Bohl J., Manning J.E. Transmission dynamics of 2019 dengue outbreak in Cambodia via next generation sequencing. *U.S.-Japan Cooperative Medical Sciences Program's 22nd International Conference On Emerging Infectious Diseases In The Pacific Rim, Bangkok, Thailand.* 2020. [poster]

32. *Swe M.M.M., Win M.M., Cohen J., Phyo A.P., **Parker D.**, Dance D., Ashley E., Smithuis F. Mapping and geographic distribution of Burkholderia pseudomallei in Myanmar. *American Society of Tropical Medicine and Hygiene*. 946. 2020. [poster]

33. *Oliveira F.L., Oristian J., Kong D., **Parker D.M.**, Leon P.V., Jeang B., Man S., Sreng S., Suon S., Huy R., Leang R., Bohl J.A., Calvo E., Valenzuela J.G., Manning J.E. Leveraging biomarkers of exposure identified in PAGODAS (Pediatric Assessment Group Of Dengue and *Aedes* Saliva) in Cambodia. *American Society of Tropical Medicine and Hygiene*. 496. 2020. [talk]

34. Yongchaitrakul S., ***Parker D.M.**, Singhasivanon P., Cui L., Sattabongkot J., Phuanukoonnon S. Understanding malaria preventive behaviors among the cross border population along the Thai-Myanmar border in Tak province, Thailand. *American Society of Tropical Medicine and Hygiene*. 757. 2020. [poster]

35. ***Parker D.M.**, Jeang B., Kong D., Man S., Sreng S., Huy R., Leang R., Bohl J.A., Valenzuela J.G., Oliveira F., Manning J.E. Individual, household, and environmental predictors of symptomatic dengue infection in a peri-urban area of Cambodia: a geostatistical analysis. *American Society of Tropical Medicine and Hygiene*. 609. 2020. [poster]

36. *Li X., Arya G.A., Thu A.M., Delmas G., **Parker D.M.**, Lwin K.M., Sriprawat K., Nosten F., Anderson T. Spatial analysis of parasite population genomics during malaria elimination efforts in Eastern Myanmar. *Genomic Epidemiology of Malaria.* 2021. [talk]

37. *Manning J., Chea S., **Parker D.M.**, Mateja A., Man S., Nhek S., Sreng S., Lay S., Fay M.P., Ly S., Sath R., Chea H., Huy R., Valenzuela J.G., Kry H., Lon C., Leang R., Oliveira F. Humoral immunity against *Aedes aegypti* saliva associated with development of inapparent dengue: a longitudinal observational cohort in Cambodia. *American Society of Tropical Medicine and Hygiene*. 126. 2021. [talk]

38. *Steinbrook E.; Min M.C.; Kajeechiwa L.; Wiladphaingern J.; Paw M.K.; Pimanpanarak M.; Hiranloetthanyakit W.; Min A.M.; Tun N.W.; Gilder M.E.; Nosten F.; McGready R., **Parker D.M.** (2021). Distance matters: barriers to antenatal care and safe childbirth in a migrant population on the Thailand-Myanmar border from 2007-2015, a pregnancy cohort study. *American Society of Tropical Medicine and Hygiene*. 0464. 2021. [talk]

39. *Bohl J.A., Lay S., Chea S., Ahyong V., **Parker D.M.**, Gallagher S., Fintzi J., Man S., Ponce A., Sreng S., Kong D., Oliveira F., Kalantar K., Tan M., Fahsbender L., Sheu J., Neff N., Detweiler A., Ly S., Sath R., Chea H., Kry H., Leang R., Huy R., Lon C., Tato C.M., DeRisi J.L., Manning J. (2021). Metagenomic next-generation sequencing of febrile patients reveals diverse vector-borne pathogen landscape in Cambodia. *American Society of Tropical Medicine and Hygiene*. 0907. 2021. [talk]

40. *Brook C., Bohl J.A., Northrup G., Lay S., Chea S., Ahyong V., **Parker D.M.**, Man S., Sreng S., Kong D., Oliveira F., Fahsbender L., Leang R., Huy R., Lon C., Tato C.M., DeRisi J.L., Boots M., Manning J. (2021). Peri-urban transmission of dengue in the Cambodian 2019 epidemic: a phylogeographic approach to understand arboviral outbreaks. *American Society of Tropical Medicine and Hygiene*. 1362. 2021. [talk]

41. *Bruckner T., **Parker D.**, Messaoudi I., Minin V. (2022). Hospital-based surveillance approximates population seroprevalence of SARS-CoV-2 in the pre-vaccine era. *Society for Epidemiological Research*. 0181. [talk]

42. *Goldstein I., **Parker D.**, Jiang S., Minin V. Using infectious disease RNA counts from wastewater surveillance to estimate the effective reproduction number. *MIDAS Network Annual Meeting*. 2022. [talk]

42. *Pacheco A.R., Lon C., Lay S., Chea S., Oum M.H., Ly S., Sath R., Tim R., Khamsiriwatchara A., Wansatid P., **Parker D.M.**, Manning J.E. The CoorFamily App: Feasibility of a 10-day location history contact tracing smartphone applicatoin for febrile patients in Cambodia. *American Society of Tropical Medicine and Hygiene*. 0692. 2022. [poster]

43. *Baker C., Barilar I., de Araujo L., Rimoin A., **Parker D.**, Boyd R., Tobias J., Moonan P.,

Click E., Finlay A., Oeltmann J., Minin V., Modongo C., Zetols N., Niemann S., Shin S. High-resolution characterization of recent tuberculosis transmission in Botswana using geospatial and genomic data - the Kopanyo study. *American Society of Tropical Medicine and Hygiene*. LB-5072. 2022. [poster]

44. *Sehi G.T., Macharia P.M., Houngbedji C.A., **Parker D.M.** A spatial analysis of geographic access to public healthcare facilities in Côte d'Ivoire. *American Society of Tropical Medicine and Hygiene*. LB-5402. 2022. [poster]

45. *Aung P.L., Soe M.T., Aung P.P., Kyaw M.P., Nguitragool W., **Parker D.M.**, Cui L., Sattabongkot J. *American Society of Tropical Medicine and Hygiene*. LB-8408. 2023. [poster]

PROFESSIONAL SERVICE

Service to the University

Department
October 2018 – June 2019
Designed and established an undergraduate minor in Global Health

December 2019 – March 2021
PhD admissions committee member (Department of Population Health and Disease Prevention)

December 2019 – June 2022
Faculty affairs committee member (Department of Population Health and Disease Prevention)

April 2020 – June 2022
Faculty workload committee member (Department of Population Health and Disease Prevention)

October 2022 – January 2023
Search committee for faculty member in Global Health (Department of Population Health and Disease Prevention)

School/Program
August 2017 – present
Director of GHREAT (Global Health Research, Education and Translation); Department of Population Health and Disease Prevention; Program in Public Health

May 2019
UCI Public Health Diverse Educational Community and Doctoral Experience Program (DECADE) Student Council, Invited Panelist, Careers Panel

August 2020 – June 2021
Search committee for Associate Dean of Research (Program in Public Health)

May 2021 – August 2023
School wide curriculum committee (Program in Public Health)

October 2021 – present
School wide Masters in Public Health advisory committee (Program in Public Health)

May 2023 – present
Co-Director of Accreditation (through the Council on Education in Public Health (CEPH))

September 2023 – present
Chair of school wide curriculum committee (Program in Public Health)

University (including UC-wide)
August 2017 – June 2023
Steering committee member GloCal Health Fellowship Program

April 2019
GHREAT-DITA Endowed Graduate Fellowship Committee

May 2019 – present
Faculty advisory board member for the UCI Infectious Diseases Initiative

Service to the field

May 2018 – May 2020
Global Assessor (reviewer for small grants); Royal Society of Tropical Medicine and Hygiene

2019
Invited expert reviewer for: European Research Council Consolidator Grants
Academy of Medical Sciences, United Kingdom, Springboard scheme

July 2020
Epidemiological investigations of Orange County Jails with regard to the COVID-19 outbreak, at request of ACLU of Southern California

August 2020
Invited ad hoc peer reviewer for: National Institutes of Health/National Institute of Allergy and Infectious Disease

2020, 2021, 2022
Invited expert reviewer for: MRC UKRI Future Leaders Fellowships

2023
Invited expert reviewer for: the Programmes Transversaux de Recherche - Institut Pasteur

2023
Invited expert reviewer for: The U.K. National Institute for Health and Care Research (NIHR) Global Health Programme

Journal Editorial Boards
August 2018 – present
Editorial board: Spatial Demography

November 2020 – present:
Associate Editor of PLOS Neglected Tropical Diseases

June 2021 – present:
Associate Editor of Discover Social Science and Health

Manuscript review for peer-reviewed journals
American Journal of Physical Anthropology, Asia Pacific Education Review, Asian Pacific Journal of Tropical Disease, BMC Infectious Diseases, BMC Medicine, BMC Public Health, BMJ Case Reports, Demography, Demographic Research, Emerging Infectious Diseases, Environmental Health Perspectives, Environmental Monitoring and Assessment, Epidemics, Epidemiology and Infection, Expert Review of Molecular Diagnostics, F1000 Research, Globalization and Health, Health and Place, Heliyon, Infectious Diseases of Poverty, International Journal of Molecular Sciences, Journal of the American Mosquito Control Association, Journal of the Association for Information Science and Technology, Journal of Mountain Science, Journal of Youth Studies, Lancet Infectious Diseases, Lancet Planetary Health, Malaria Journal, Pathogens and Global Health, PLoS One, PLoS Medicine, Population and Environment, Scientific Reports, Social Science & Medicine, Southern Economic Journal, Spatial Demography

TEACHING/ADVISING

University of California, Irvine (UGs = undergraduate students; Gs = graduate students)

Term	***Course number: course name (% contribution)***	***Student count***
2018 (Spring)	PH101: Introduction to Epidemiology (100%)	246 UGs
2018 (Fall)	PH213: International Epidemiology (100%)	13 Gs
2019 (Spring)	PH101: Introduction to Epidemiology (100%)	242 UGs
2019 (Fall)	PH213: International Epidemiology (100%)	8 Gs
2020 (Spring)	PH101: Introduction to Epidemiology (100%)	250 UGs
2020 (Spring)	PH289: Climate Change and Global Health (100%)	14 Gs
2020 (Fall)	PH213: Epidemiology in Global Health (100%)	25 Gs
2021 (Spring)	PH101: Introduction to Epidemiology (100%)	247 UGs
2021 (Fall)	PH213: Epidemiology in Global Health (100%)	26 Gs
2022 (Spring)	PH101: Introduction to Epidemiology (100%)	252 UGs
2022 (Fall)	PH213: Epidemiology in Global Health (100%)	30 Gs
2023 (Spring)	PH282: Climate Change and Global Health (100%)	10 Gs
2023 (Fall)	PH213: Epidemiology in Global Health (100%)	33 Gs

Mahidol University, Bangkok, Thailand
2021 (Fall) TMHG548: Research Methods in Health Informatics (20%)

Graduate Student Committee Member (all from UC-Irvine Public Health unless otherwise indicated)

Current PhD (pre-candidacy)
09/2021 – present Heven Sime, Co-Advisor; Addis Ababa University
11/2021 – present Prudencio Merino, Committee member; Lundquist Institute

	for Biomedical Innovation at Harbor-UCLA Medical Center
09/2022 – present	Marcela Lopez, Co-advisor
09/2023 – present	Jose Villalvazo, Co-advisor
09/2023 – present	Adriana Urrutia, Advisor

Advanced to PhD Candidacy

09/2020 – present	Natasha Glendening, Advisor
07/2021 – present	Chelsea Baker, Committee member; Nursing
09/2021 – present	Gaëlle Sehi, Advisor
09/2021 – present	Maia Tarnas, Advisor
09/2023 - present	Catalina Medina, Committee member; Statistics

Completed PhD

12/2017 – 06/2018	Tamara Jimah, Committee member
08/2017 – 06/2019	Sayambhu Saita, Chair and external committee member; Mahidol University, Thailand
05/2019 – 06/2021	Kasama Pooseesod, Chair and external committee member, Mahidol University, Thailand
12/2019 – 01/2020	Alfonso Shin, Committee member; Biomedical Engineering
09/2021 – 08/2023	Wen-Jian Chung, Committee member; Computational Science (UCI and UCSD joint program)
2022	Desmond Gul, External PhD thesis examiner, Monash University, Australia
01/2022 – 05/2022	Samuel Lopez, Committee member; Mathematics
05/2022 – 06/2023	Damon Bayer, Committee member; Statistics
04/2019 – 12/2023	Lauren Bradley, Committee member; Biological Sciences

Current MPH (Masters in Public Health)

01/2023 – present	Victoria Sahrai, MPH Advisor

Completed MPH/MS

09/2019 – 06/2021	Alexanhder Luu Vo, MPH Advisor
09/2019 – 06/2021	Sasha Jahan Belaluddin, MPH Advisor
05/2021 – 01/2022	Tullaya Sitasuwan, MS, External Thesis Chair, Mahidol University, Thailand
09/2021 – 06/2022	Jay Mantuhac, MPH Advisor
09/2022 – 06/2023	Rohan Subramanian, MPH Advisor
01/2023 – 06/2023	Shemms Najjar, MPH practicum site supervisor
12/2021 – 09/2023	Kamaori Xiong, MPH Advisor

Undergraduate Research Advising

2018/2019 Stephanie Nguyen and Nhu Do: Psychosociocultural investigation of smoking cessation of previous Vietnamese male smokers in Orange County (*awarded $400 for research; presented at UROP symposium Spring 2019*)

2018/2019 Elizabeth Savin, Jessica Oshiro, Noor Elbershawi, Rita L. Karaian and Mahima Avanti: Flu vaccine perceptions among University of California, Irvine (UCI) students (*awarded $648 for research; presented at UROP symposium Spring 2019*)

2017/2018 Samar Arshad and Catherine Chiou: Assessing the knowledge of and access to contraceptives among female undergraduate students at the University of California, Irvine (*awarded $500 for research; presented at UROP symposium Spring 2018*)

2019/2020 Hector M. Garcia: Vaccines: Role of a UCI student's living situation in vaccine barriers (*awarded $350 for research on access to the influenza vaccine among UCI students*)

2019/2020 Chas P. Hurst, Nydia I. Gomez, Aqsa Khalil: Exploring food insecurity and its effect on academics and wellbeing among college students (*awarded $400 for research on food security among UCI students*)

2020/2021 Chas P. Hurst: Examining Potential Associations between Landscape Attributes, Environmental Factors, *Aedes* Mosquito Abundance, and Human Exposure to *Aedes* Mosquitoes in Cambodia *(awarded $600)*

2020/2021 Catthi Ly, Aisha Raheem, Nydia Gomez, Jasmine Quan, Thang Nguyen, Jihwan T. Kim, Govena Garewal, Chas P. Hurst, Katy X. Dang, Sarah Wang, Nayeli Inzunza, Manuel Sardaryan, Jayden Ware, Samuel D. Tincher, Ashley Park, Chisomam J. Agaranna: Community Based Participatory Research Project for Orange County amid the COVID-19 Pandemic *(awarded $2,500)*

2021/2022 Nydia I. Gomez, Shayla Tran, Annaliese Kaxon-Rupp, Ashley Tang, Jasmine Quan, Ziena Smadi, Ashley Park, Chisomam Agaranna, Benjamin Ho, Dana Suleiman, Melyssa Karcher: Assessing the knowledge of sexual health and sexual health resource use on UCI undergraduate students and its relation to cultural upbringing (*awarded $200*)

Workshops

2014 Workshop on mapping health structures for malaria elimination. Community Partners International, Karen Department of Health and Welfare, Karen Baptist Convention, Dawei, Myanmar; October 27 – 29.

2015 Introduction to global positioning systems and geographic information systems for malaria elimination. Save the Children Nepal and Epidemiology and Disease Control Division, Department of Health Services, Hetauda and Chitwan, Nepal; December 8 – 9; Kathmandu, Nepal; December 11.

2016 Introduction to ArcMap and GIS for malaria elimination. Save the Children Nepal and Epidemiology and Disease Control Division, Department of Health Services, Kathmandu, Nepal; June 16.

2016 Introduction to Quantum GIS (QGIS). Lao-Oxford-Mahosot Hospital Wellcome Trust Research Unit, Vientiane, Laos; May 23 – 25.

2016 Mapping health services in Eastern Burma Workshop. Burmese Medical Association and Karen Department of Health and Welfare, Mae Sot, Thailand; March 30 – April 1 (with Saw Poe Wah and Saw Myo Chit Min).

2016 Mass drug administration in low-income countries. University of Copenhagen, School of Global Health; July 25 – August26. Contributed five short videos on mass drug administration for malaria.

2017 Introduction to Quantum GIS (QGIS). Shoklo Malaria Research Unit, Mahidol Oxford Tropical Medicine Research Unit, Mae Sot, Thailand; June 17 – 18.

2020 Health Equity Contact Tracing. University of California, Irvine; July 20 – September 18.

2022 UC Irvine Summer Institute in Biostatistics and Undergraduate Data Science; Participated as a data science/infectious disease epidemiology mentor

PROFESSIONAL MEMBERSHIPS (*active)

American Association for the Advancement of Science*
American Association of Physical Anthropologists
American Public Health Association*
American Society for Rickettsiology*
American Society of Tropical Medicine and Hygiene*
International Society for Infectious Diseases*
John Snow Society*
Population Association of America
Royal Geographic Society*
Royal Society of Tropical Medicine and Hygiene
Sigma Xi*

MEDIA & OTHER INTERVIEWS

Media about my research

April 2015: Interviewed by Alexa C. Kurzius for article titled “Mapping Malaria in Myanmar” on the Johns Hopkins Bloomberg School of Public Health “Global Health Now” blog: available at: https://www.globalhealthnow.org/2015-04/mapping-malaria-myanmar

March 2016: Interviewed by Sarika Bansal for article titled “The Unrelenting Specter of Drug-Resistant Malaria” on “The Development Set”: https://medium.com/the-development-set/the-unrelenting-specter-of-drug-resistant-malaria-da11f736c973#.oypacg437

April 2019: Article about my research on herd effects from mass administration of antimalarials: https://www.drugdiscoverytrends.com/mass-drug-administrations-can-grant-population-protection-against-malaria/

December 2019: Interview with Lilibeth Garcia about my malaria work for UCI's webpage: https://news.uci.edu/2020/01/13/on-a-mission-to-mitigate-malaria/

August 2020: Article about our online dashboard for local COVID-19 epidemiology: https://www.audacy.com/knxnews/articles/uci-launches-covid-19-trends-site-comparing-socal-counties

August 24, 2021: Interviewed by Ben Brazil from LA Times OC about my research on the spatial epidemiology of COVID-19 in Orange County: https://www.latimes.com/socal/daily-pilot/entertainment/story/2021-08-26/uci-disparities-orange-county-covid-19

August 23, 2021: Quoted in the OC Business Journal with regard to our paper on the spatial epidemiology of COVID-19 in OC: https://www.ocbj.com/news/2021/sep/06/uci-finds-disparities-covid-infections-deaths/

September 2021: Article about our paper on Rickettsial diseases, climate, and landscape changes in Lao PDR: https://www.news-medical.net/news/20210918/Changing-weather-patterns-in-Laos-linked-to-increase-in-typhus-cases.aspx

Interviews in my areas of expertise

March 2, 2020: Interviewed by Alicia Lee from CNN about hand sanitizer: https://www.cnn.com/2020/03/03/health/coronavirus-diy-hand-sanitizer-trnd/index.html

March 3, 2020: Interviewed by Patrice Apodaca of Daily Pilot about hand hygiene: https://www.latimes.com/socal/daily-pilot/opinion/story/2020-03-10/coronavirus-one-reason-to-stop-shaking-hands

March 20, 2020: Interviewed by Liz Scherer about hand hygiene: https://www.everydayhealth.com/coronavirus/should-you-diy-hand-sanitizer-health-experts-weigh-in/

May 1, 2020: My expert opinion was referenced by ACLU of Southern California regarding COVID-19 in jails: https://www.latimes.com/california/story/2020-05-01/warning-coronavirus-could-quadruple-in-orange-county-jails-aclu-sues-oc-sheriff-to-free-inmates

May 4, 2020: Interviewed by Katie Murar for OC Business Journal regarding COVID-19 in Orange County: https://www.ocbj.com/news/2020/may/04/covid-19-first-coast-now-inland/

May 15, 2020: Interview with Paul Anderson of the City News Service regarding COVID-19 in Orange County: https://www.nbclosangeles.com/news/coronavirus/orange-county-reports-155-more-cases-of-covid-19/2364155/

May 18, 2020: Interviewed by Ashley Ludwig regarding COVID-19 in Orange County https://patch.com/california/orange-county/nearly-4-500-cases-coronavirus-orange-county

July 22, 2020: Interviewed by Jessica De Nova for ABC7 regarding geographic spread of COVID-19 in OC: https://abc7.com/orange-county-coronavirus-oc-covid-19-santa-ana/6329876/

July 28, 2020: Interviewed by Tyler Kingkade regarding contact tracing for COVID-19 in California: https://www.nbcnews.com/news/us-news/california-coronavirus-cases-spike-contact-tracing-stalled-fear-embarrassment-n1235345

August 15, 2020: Interviewed by Tammy Webber for an article on fear of contact tracing among immigrants in the Associated Press: https://apnews.com/67d8c0778472c9c06bb5b1a0622375c2

November 6, 2020: Interviewed by Spencer Custodio for an article about COVID-19, risk behaviors, and a potential second peak in the Voice of OC: https://voiceofoc.org/2020/11/is-a-second-coronavirus-wave-hitting-orange-county/ and: https://voiceofoc.org/2020/11/second-coronavirus-wave-hits-orange-county/

December 9, 2020: Interviewed by Vera Castaneda for the LA Times Daily Pilot about our seroprevalence survey in Santa Ana, California: https://www.latimes.com/socal/daily-pilot/entertainment/story/2020-12-10/drive-through-and-walk-in-coronavirus-antibody-testing-launches-in-santa-ana?_amp=true

January 26, 2021: Quoted in USA Today about naming conventions for different variants of SARS-CoV-2: https://www.usatoday.com/story/news/health/2021/01/26/new-covid-variants-brazil-uk-south-africa-explained/4257748001/

March 18 and 19, 2021: Quoted in the Voice of OC with regard to community efforts to reduce COVID-19 transmission and to focus vaccinations on places with the heaviest COVID-19 burden: https://voiceofoc.org/2021/03/community-efforts-to-bring-covid-vaccines-to-ocs-hardest-hit-neighborhoods-are-paying-off/ and https://voiceofoc.org/2021/03/covid-19-year-one-ocs-doctors-nurses-community-organizations-stepped-up-to-curb-the-virus/

May 3, 2021: Quoted in the Orange County Register about incomplete vaccination coverage data and missing important parts of our population with regard to vaccination: https://www.ocregister.com/2021/05/03/ocs-vaccination-efforts-shift-from-the-vulnerable-to-the-hesitant/

May 19, 2021: Quoted in the Voice of OC with regard to the CDC's changing recommendations for masking after vaccination: https://voiceofoc.org/2021/05/fully-vaccinated-orange-county-residents-still-required-to-wear-masks-indoors-for-another-month/

June 7, 2021: Interviewed by Marilynn Larkin for Reuters with regard to co-infection by SARS-CoV-2 and Helminths: https://dailydivafinds.com/covid-19-helminth-syndemic-may-curb-vaccine-efficacy-in-tropical-areas/

July 9, 2021: Quoted in Voice of OC with regard to local businesses dealing with reopening, masks, and vaccine verification: https://voiceofoc.org/2021/07/oc-residents-and-businesses-grapple-with-masks-vaccine-verification-policies-amid-coronavirus-reopening/

December 22, 2021: Quoted in the Voice of OC with regard to schools as important community partners for COVID-19 public health efforts: https://voiceofoc.org/2021/12/state-and-local-health-experts-move-to-vaccinate-kids-ahead-of-expected-winter-surge/

January 6, 2022: Quoted in the Voice of OC with regard to accessing COVID-19 testing in Orange County: https://voiceofoc.org/2022/01/school-absences-anger-in-oc-as-hunt-for-covid-19-tests-turns-up-fruitless-for-many/

February 11, 2022: Quoted in Business Insider with regard to mask mandates for COVID-19: https://www.businessinsider.com/mask-mandates-lifted-states-reasonable-experts-2022-2

March 18, 2022: Quoted in U.S.A. Today with regard to COVID-19 variants and concerns about new waves of the disease: https://www.usatoday.com/story/news/health/2022/03/18/omicron-variant-covid-19-coronavirus/7089686001/

April 7, 2022: Quoted in an article on Healthline with regard to COVID-19 variants and current epidemiological trends: https://www.healthline.com/health-news/why-covid-19-cases-may-continue-to-decline-in-the-united-states

April 21, 2022: Quoted in USA Today with regard to masking after mask mandates have been dropped for public transport: https://www.usatoday.com/story/travel/news/2022/04/21/mask-mandate-face-mask-travel/7381589001/

May 1, 2022: Quoted in the Wall Street Journal with regard to SARS-CoV-2 Omicron subvariants: https://www.wsj.com/articles/covid-19-cases-rise-in-the-u-s-with-limited-impact-11651397401

January 9, 2023: Quoted in the Orange County Register with regard to geographic and demographic variation in SARS-CoV-2 infection outcomes: https://www.ocregister.com/2023/01/08/why-do-some-in-orange-county-die-from-covid-19-but-others-don't/

March 24, 2023: Quoted in Parents with regard to Babesiosis (a tick-borne disease): https://www.parents.com/tick-borne-illness-babesiosis-7370679

April 20, 2023: Quoted in the Orange County Register with regard to *Aedes* mosquitoes and the risk of *Aedes*-borne disease in Southern California: https://www.ocregister.com/2023/04/19/mosquito-season-just-got-worse-heres-what-you-need-to-know/

June 29, 2023: Quoted in HealthLine with regard to local malaria transmission in the U.S.A.: https://www.healthline.com/health-news/locally-transmitted-malaria-detected-in-u-s-for-first-time-in-20-years?utm_source=ReadNext

July 7, 2023: Quoted by CNBC with regard to local transmission of malaria in the U.S.A.: https://www.cnbc.com/2023/07/07/us-malaria-risk-low-despite-local-cases-what-you-need-to-know.html

August 4, 2023: Quoted in HealthLine with regard to rising cases of leprosy in Florida: https://www.healthline.com/health-news/leprosy-endemic-florida-what-to-know

August 18, 2023: Quoted in Parade with regard to a new SARS-CoV-2 variant: https://parade.com/health/eris-eg5-new-covid-variant